Isaiah Nyakundi

Data Engineer | Python, SQL, ETL / ELT pipelines

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PROFILE

Data engineer with 2+ years of freelance experience and four public portfolio projects covering incremental ingestion, distributed processing, orchestration and data-quality validation. Background in biostatistics; works from the data consumer's requirement back to a tested, documented, rerunnable pipeline.

TECHNICAL SKILLS

Languages: Python, SQL, Spark SQL
Processing & storage: Apache Spark, PySpark, PostgreSQL, Parquet, Delta Lake
Orchestration & modelling: Apache Airflow, dbt, incremental loading with watermarks, medallion architecture
Platforms & delivery: Databricks, Snowflake (design), Azure ADLS Gen2 (design), Docker, Git/GitHub, pytest
Learning: Microsoft Fabric

EXPERIENCE

Freelance Data Engineer

Client engagements | 2+ years | client details confidential

- Build data extraction, transformation and validation workflows that deliver analytics-ready datasets to clients.
- Translate requirements into pipeline steps and architecture decisions, prioritising maintainability, data quality and repeatable execution.
- Document data flow and implementation choices so the work can be handed over and maintained.

PORTFOLIO PROJECTS (SELF-DIRECTED, PUBLIC CODE)

E-Commerce Data Platform

Airflow, dbt, PostgreSQL, Docker, pytest | github.com/viradidi/ecommerce-data-platform

- Airflow DAG (extract, validate, load, dbt build) scheduled daily; loads upsert on the business key so reruns do not duplicate.
- Validation gate fails the run on null keys, duplicates and negative amounts; 6 pytest tests and dbt not_null/unique tests.
- Runs in Docker Compose (PostgreSQL, Redis, Airflow with Celery). Built on sample data.

Novacart: Incremental Data Pipeline

Databricks, PySpark, Delta Lake | github.com/viradidi/Novacart-casestudy

- Bronze/Silver/Gold notebooks with a composite (timestamp, key) watermark and a control table per layer recording position, run ID and row counts.
- Silver standardises, keeps the latest record per key, merges by key and quarantines invalid rows; Gold recomputes only impacted orders and keeps SCD Type 2 history.
- No automated tests or orchestration yet.

Retail Analytics Data Pipeline

PySpark 3.5.1, Docker, Parquet | github.com/viradidi/retail-pyspark

- Medallion pipeline over a synthetic retail dataset (generator configured for 1,000,000 records) seeded with invalid values.
- Silver rules nullify impossible ages, quantities, prices, discounts and ship dates; Gold produces daily sales and city revenue metrics in Parquet.
- Spark standalone cluster (1 master, 2 workers) in Docker Compose.

FlowBridge: Supply Chain Data Platform

Snowflake, Azure, SQL, Python | github.com/viradidi/FLOWBRIDGE-SNOWFLAKE-PROJECT

- Documented design: Azure ADLS Gen2 with Event Grid and Snowpipe into Snowflake Bronze/Silver/Gold layers, star schema, RBAC and secure views.
- Architecture and business documentation published; SQL implementation scripts are not yet committed.

EDUCATION

Bachelor of Science in Biostatistics (final year)
Jomo Kenyatta University of Agriculture and Technology (JKUAT), Kenya